

Sustainable water pumping using hybrid renewable energy sources

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Abstract. The water-energy nexus indicates today's intricate interdependence of water and energy and holds the key to human well-being and national development. It is particularly significant in countries with arid climatic conditions, such as is the case with Namibia, where water scarcity is coupled with a huge gap between electricity supply and demand. In this paper, we assessed the viability of using Hybrid Renewable Energy Systems (HRES) to sustainably power borehole schemes in Namibia. As a case study, grid-connected photovoltaic (PV), onshore wind, and battery storage technologies were used to meet the electric energy demands at the Berg Aukas borehole scheme, which consumes an average of 7,820 kWh/day with a peak demand of 1,162.5 kW. HOMER Grid software was used to model the utility grid tariff and net metering structures and determine optimal system configurations. For the sensitivity analysis, the discount rate (interest rate) was used as a sensitivity variable. The results show that at Namibia's average discount rate of 6.7%, the proposed optimal HRES had a levelized cost of energy (LCOE) and payback period of USD 0.0929 /kWh and 4.17 years, respectively. At a discount rate of 10%, the optimal HRES had an LCOE of USD 0.105 /kWh with a projected payback period of 3.77 years. At a discount rate of 13%, the LCOE was USD 0.115 /kWh with a payback period of 3.43 years. For all the modelled instances, the proposed systems for the borehole scheme were found to be economically feasible as each had a LCOE that was lower than that of the utility company (CENORED), i.e., USD 0.283 /kWh.

Keywords: HRES; HOMER Grid; optimization; sensitivity analysis; capacity factor; discount rate; payback period; LCOE

1 Introduction

With the arid climatic conditions, most parts of Namibia have very little surface water. Over the years, this has resulted in the increased adoption and popularity of groundwater abstraction from aquifers via boreholes to satisfy the water needs of communities. At most of the existing borehole schemes in Namibia, pump installations are powered by small wind turbines (usually off-grid), electricity from the grid, diesel engines or manual hand pump

technologies (usually seen in rural settings) [1]. The Namibia Water Corporation Ltd (NamWater) is the government-owned utility responsible for the bulk supply of water to industries, government institutions, municipalities, and commercial entities such as mines. Although it is not privatized, NamWater is a commercialized water supply entity and hence must recover the full cost of water supply to provide its customers reliably and sustainably with water. According to NamWater's annual reports, their greatest expenditure is electricity charges paid to electric utility companies. This in turn influences the cost of supplied water, and hence the energy charges paid by NamWater are eventually passed on to end-consumers. The main aim of the study was to assess the techno-economic feasibility and sustainability of using hybrid renewable energy systems (HRES) to power borehole pump stations in Namibia. We therefore designed and simulated a HRES, using Hybrid Optimization of Multiple Energy Resources (HOMER) software, for a selected borehole in Namibia. We then performed quantitative analysis of the hybrid system to determine the involved costs and viability of implementing a hybrid groundwater pump at the specified borehole. In section 2 the case study borehole scheme is described. In section 3 the methods and materials used is presented. In section 4 the results are presented and discussed. In section 5 conclusions are drawn and recommendations for further work are proffered.

2 Berg Aukas Borehole Scheme

The Berg Aukas borehole scheme was selected to be used as a case study. Its coordinates are -19.5131874° , 18.2459092° , latitude and longitude, respectively. It is a groundwater scheme comprising of two boreholes near the decommissioned Berg Aukas Mine and a pipeline network, which supplies potable water to the Berg Aukas Mine settlement, the Otijituo and Elandspan communities, situated in the communal part of eastern Otjozondjupa region. According to a 2020 Environmental Management Plan (EMP) prepared by NamWater, $514\,183\text{m}^3$ of water was sold from this scheme [6].

The groundwater at the Berg Aukas scheme is abstracted from two production boreholes and pumped to a concrete ground level collector reservoir, which is situated to the north of the decommissioned Berg Aukas Mine [6]. These two boreholes are labelled WW2153 and WW24857 (see also Figure 1) and are used by NamWater to draw water from the GTO-WCA Karst aquifer that lays underneath the area.

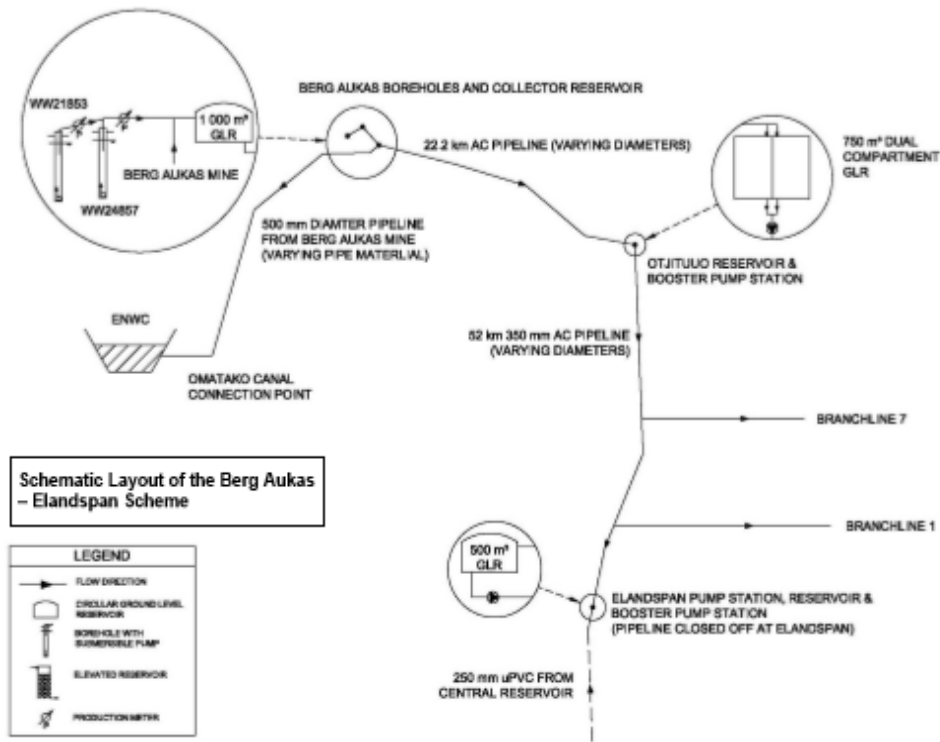


Fig. 1. Berg Aukas borehole scheme schematic (Source: <http://the-eis.com/elibrary>)

Submersible pumps are installed and manually operated at each of these two boreholes. It is these pumps that abstract groundwater from the aquifer and into the Berg Aukas Collector Reservoir. These pumps are operated in parallel [2]. The main power supply to the pump installations is via a dedicated 11 000/400 V, 100kVA pole-mounted transformer situated inside the pump installation yards; the pump installations themselves are supplied via a 100 A TP circuit breaker [2]; this was the motivating factor to design a grid-connected hybrid system, as opposed to an off-grid system.

3 Method and Materials

3.1 Methodology Flowchart

The methodology adopted involves defining inputs to HOMER, such site-specific metrological data, the load profile, economics data, and the HRES components/technologies to be used. HOMER then performs the system simulation, optimisations and sensitivity analysis as indicated in the following subsections. The results of the optimal HRES sizing, performance details and all financial computations are subsequently collated.

3.2 Meteorological Data for Berg Aukas

The annual daily average solar radiation for the Berg Aukas area was determined to be 5.82 kW/m²/day (as per the NREL database). This region, as with most of Namibia, has abundant solar potential and is typically cloudless throughout the year. The wind speed data was obtained from the NASA meteorological database. The annual average wind speed is 5.42m/s. Multiple studies have found that areas with an annual average wind speed greater than 4.5m/s are able to leverage onshore wind as a source of energy [3]. The meteorological data (solar insolation, wind speed, ambient temperature) were obtained from the National Meteorological Centre, Luzira, Kampala. The data were captured at the automatic weather station located in Tororo, which is the nearest weather station to Busia district.

3.3 The Electrical Load

HOMER Grid allows designers to define/upload the hourly, time-series electrical load data that the hybrid system is expected to serve. These loads represent the electrical demands the HRES must be able to meet at each time-step (hourly). Unfortunately, attempts at acquiring measured load data from the operators of the borehole scheme were unsuccessful. An alternative approach to model load demands had to be adopted.

The groundwater that is abstracted from the Berg Aukas borehole scheme supplies potable water to the Berg Aukas Mine settlement, and the Otijituuo and Elandspan communities [2]. The water demand for these communities varies seasonally (summer months higher the winter months). A study conducted by Wang L, Kinzelbach W, Yao H et al focused on determining the accuracy of using electric meter readings for indirect groundwater abstraction monitoring and noted that the most important proxy for groundwater use is energy consumption, which is a good approximation proportional to water abstraction [4]. Their results showed that energy-based groundwater abstraction monitoring outperforms direct water-metering readings with respect to cost and robustness and had a mean error rate of 5%. This was theoretical basis upon which the approach used to model the expected load demands for the Berg Aukas borehole scheme was developed.

HOMER allows designers to use synthetic load profiles to generate realistic approximations of load demands the HRES is expected to meet on an hourly basis. Without access to NamWater's hourly electrical load data for the Berg Aukas borehole scheme, data from the borehole scheme's electricity supplier, CENORED, was used. According to CENORED's annual reports, the average energy-consumed at the Berg Aukas pump station in a year is approximately 2,854,325 kWh.

The annual, average daily consumption was determined as, $E_{av-daily} = \frac{2,854,325 \text{ kWh}}{365} = 7,820 \text{ kWh/day}$

In other words, over the course of a year, the per day consumption for the borehole pumpstation was 7820 kWh/day; this value was used as the scaled annual average in HOMER.

The 24-hour load profiles designers can choose from in HOMER are residential, commercial, industrial and community. The daily load profile that was selected for the Berg Aukas scheme was that of a community. As noted earlier, the studies conducted by Wang L, Kinzelbach W, Yao H et al. established the accuracy of using energy consumption as a proxy for groundwater use. From this, it would be reasonable to assume that the pattern of electric energy consumption at the borehole scheme, over the course of day, would closely mimic that of the water demands of the communities being supplied water from the borehole

For the synthetic load to have a cyclic and seasonal annual variation, January was selected as a peak month because this would incorporate increased water demand and, by extension, energy consumption, during the Namibian summer periods, into the load profile. The generated (synthetic) seasonal and daily load profiles for the Berg Aukas borehole scheme are shown in Fig. 2 and Fig. 3. .

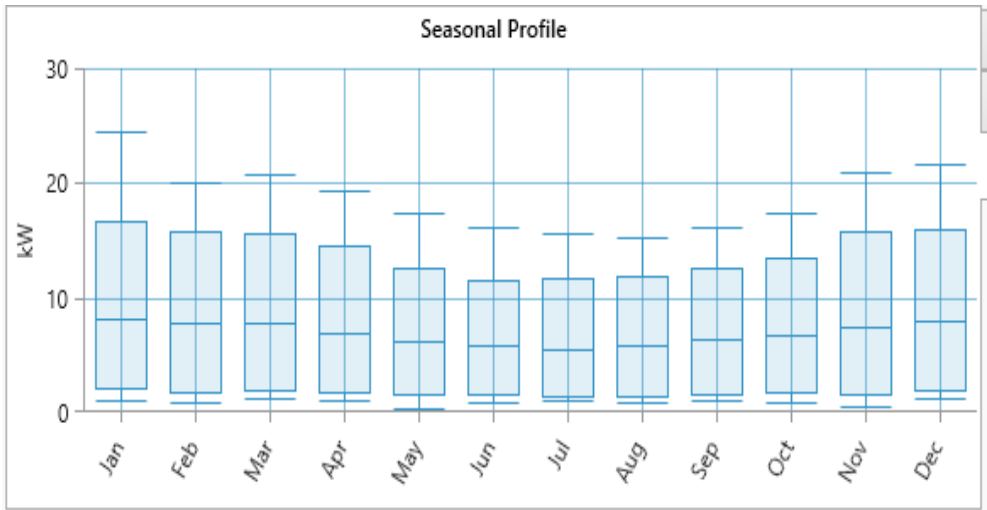


Fig. 2. Seasonal load profile for the Berg Aukas borehole scheme using synthetic load data

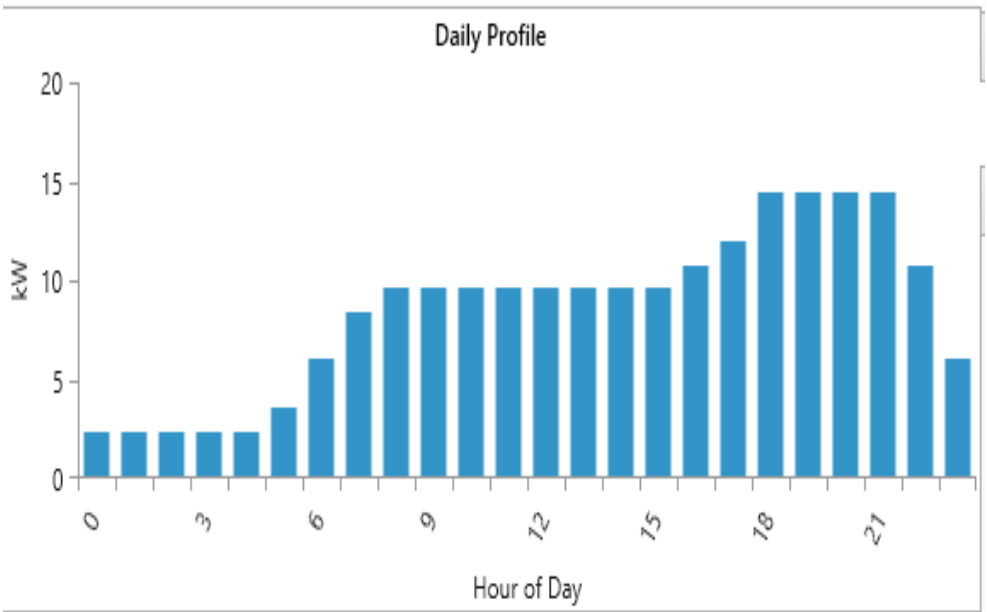


Fig. 3. Daily load profile (community) for the Berg Aukas borehole scheme using synthetic load data

From the synthetic load, the annual energy consumption was 2,854,325 kWh, with an

average of 7,820 kWh/day and peak demand of 1,162.5 kW.

3.4 System Components

Fig. 4is the general configuration for the hybrid renewable energy system used in this study.

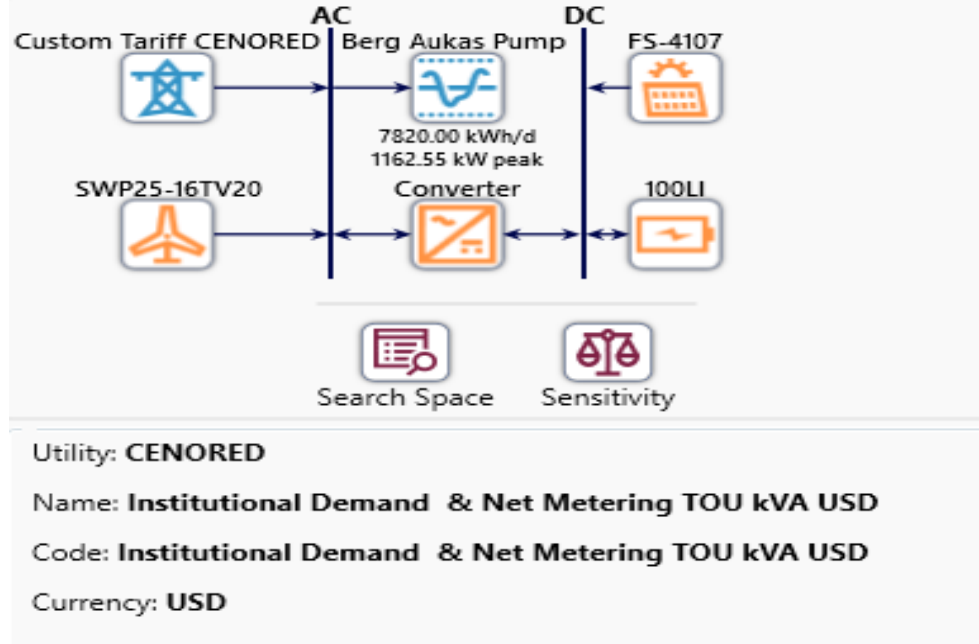


Fig. 4. General configuration of proposed on-grid HRES

The technical properties of the selected system components are given in Table 1.

Table 1: Technical properties of system components

Component	Parameter	Value
Flat plate photovoltaic panel (First Solar Series 4 – FS 4107)	Nominal Power ($\pm 5\%$)	107.5 W
	Nominal operation cell temperature	45°C
	Temperature coefficient	-0.34%/ °C
	Efficiency at standard test conditions	14.9%
	Derating factor	88%
	Open circuit voltage	86.6V
	Short circuit current	1.75A
	Voltage at maximum power	69.4V
	Current at maximum power	1.59A
System Converter	Efficiency	95%

Generic 100kWh Li-ion Battery	Nominal Voltage (V)	600V
	Nominal Capacity (kWh)	100 kWh
	Maximum charge current (A)	167V
	Maximum discharge current (A)	500A
	Throughput	300,000kWh
SWP25-16 wind turbine	Rated capacity	25kW
	Rotor diameter	16m
	Number of blades	3
	Hub height	30m
	Rated wind speed	11m/s
	Cut-in wind speed	3m/s
	Cut-out wind speed	25m/s
	Extreme wind speed	52.5m/s

3.5 Component Costs

The costs directly associated with renewable energy systems are the total installed cost, replacement cost, and operation and maintenance (O&M). The values listed in Table 2 were extracted from the International Renewable Energy Agency (IRENA) renewable cost database [5].

Table 2: Energy type and costs of system components

Energy type/component	Lifetime (Years)	Total Installed Cost (USD/kW)	Replacement cost (USD/kW)	O&M cost
PV solar	25	1,575	1,400	USD 9.0/kW/yr
System converter	20	600	600	USD 0/kW/yr
Battery Storage	15	700	700	USD 10/kW/yr
Onshore wind	25	1873	1800	USD 56/kW/yr

These values are the weighted-average costs of documented renewable energy technologies and projects in South Africa. Because of the similarities between the Namibian and South African marketplace, these values were assumed to be reasonably credible. The decision to include battery storage in the proposed on-grid HRES was motivated by the current net metering policy of Namibia. The ECB has set limits to the energy net metered consumers are allowed to feed back into the grid [6]. The inclusion of an energy storage medium is beneficial when there is excess electric energy production, and the net metering limit has been reached. The inclusion of battery storage also improves the overall reliability of the system.

3.6 Economics

HOMER requires designers to define the nominal discount rate, expected inflation rate and the project lifetime. Note that the nominal discount rate, commonly referred to as the nominal interest rate, is computed using eqn. (1).

$$i = \frac{i' - f}{1 + f}$$

(1)

where,

- i : real discount rate (real interest rate)
- i' : discount rate (nominal interest rate)
- f : expected inflation rate

Namibia’s nominal discount rate, i.e., interest rate, averaged 6.70 % between 2007 and 2021. The expected inflation rates for 2021, 2022, and 2023 are 3.44%, 4.44% and 4.5%, respectively, which averaged to 4.12%. The project lifetime used in the simulations was 25 years.

4 Results and Discussion

4.1 Consideration

The general system configuration that was fed into the simulation engine composed of solar PV, onshore wind, an energy storage medium, the grid, and a system converter. HOMER was then tasked with figuring out what component sizing would result in the lowest net present cost (over the 25-year project lifetime), while simultaneously being able to meet the electric load demands of the pumpstation. According to the International Renewable Energy Agency (IRENA), both the capacity factor and cost of capital have a significant impact on the Levelized Cost of Energy (LCOE) of renewable power projects [5]. The sensitivity variable for the sensitivity analysis of this HRES study was the nominal discount rate because it directly influences the cost of capital. Hence, the system configuration and component sizing were optimized at discount rates of 6.7% (Namibia’s average discount rate), 10% and 13%. This was important because it showcased the influence discount rates have on the system configuration, sizing, payback period and ultimately the Levelized Cost of Energy (LCOE). The currency used in the tables and figures below was USD (with an exchange rate of 15.11 NAD/USD)

4.2 Discount Rate of 6.7%

At this discount rate, the optimal system configuration consisted of the grid, solar PV, battery storage and onshore wind.

4.2.1 System Sizing

As with any constrained optimization problem, there must be an objective function along with its constraints. HOMER’s objective function was to minimize the proposed HRES’s net present cost (NPC) while still being able to meet the borehole scheme’s electric load

requirements. The optimal component sizing at this discount rate was found to be as shown in Table 3.

Table 3: System components with their optimal sizes at a discount rate of 6.7%

Component	Sizing
Grid	N/A
Solar PV (FS-4107)	222 kW
Battery Storage (Generic 100kWh lithium-ion battery)	2 units (200kWh of storage capacity)
Converter	148kW
Wind turbine (SWP25-16TV20)	975kW

4.2.2 Performance Details

The borehole scheme consumes 7826kWh/day and has a peak of 1163 kW. In the proposed optimal HRES for a 6.7% discount rate, the proportion of electric energy generation sources that served the load are illustrated in Fig. 5. .

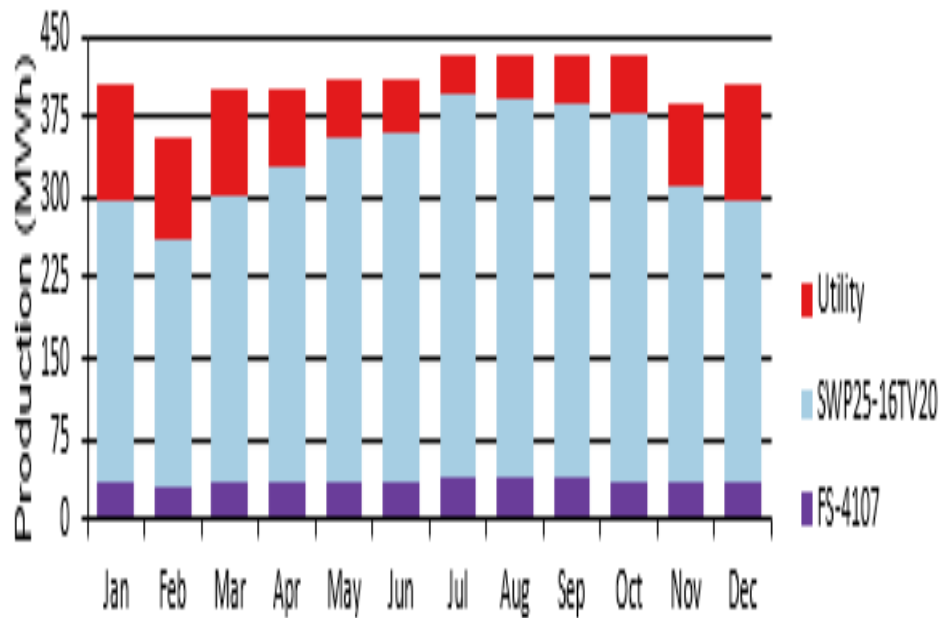


Fig. 5. Electric energy sources for proposed HRES at 6.7% discount rate

The renewable fraction is the percentage of the energy delivered to the load that originated from renewable power sources [7]. The renewable fraction for this proposed HRES is 80.6%. This is not to be confused with the proportion of energy produced from renewable sources which was 82.83% in this case. In grid-connected systems, the renewable

fraction will typically be less than the total proportion of energy produced from renewable sources. This difference is attributed to the fact that some of the energy produced by the renewable conversion devices will go towards charging the battery storage and net-metering. Table 4 shows the component-wise annual energy production at 6.7% discount rate.

Table 4: Annual energy production per component for the HRES at 6.7% discount rate

Production	kWh/yr	Percentage (%)
First Solar Series 4-107	413,862	8.43
SWP25-16TV20 (inverter version)	3,653,467	74.4
Grid Purchases	841,026	17.1
Total	4,908,355	100

4.2.3 Financials

At this discount rate the initial capital investment of the proposed optimal HRES was USD 2.4million with a projected simple payback period of 4.17 years. The most important metric for comparison between the base case (grid-only) and the recommended HRES is the levelized cost of energy (LCOE). The proposed system’s expected LCOE was USD 0.0929 /kWh, which is lower than that of the grid (USD 0.283/kWh). The corresponding economic metrics at the 6.7% discount rate are:

- Simple payback period: 4.17 years
- Return on Investment: 18.3%
- Internal Rate of Return: 23.1%
- Net Present Value: \$7.47 million
- Capital Investment: \$2.40 million

The annualized savings is the difference in annualized cost between the grid-connected (base case) system and the proposed HRES system. Annualized cost is the cost that, if it were to repeat in every year over the project lifetime, would give the same net present cost as the actual cash flow sequences [7]. Table 5 highlights the annual utility bills for both the base case and the proposed system at a 6.7% discount rate, it is divided into the three main parts of a typical utility bill, namely, consumption, demand, and fixed charges.

Table 5: Annual utility bills and savings by category, for HRES at 6.7% discount rate

	Consumption Charge	Demand Charge	Fixed Rate	Total
Base Case	\$473,335	\$332,891	\$1,517	\$807,743
Proposed Case	-\$87,237	\$258,455	\$1,517	\$172,735
Annual Savings	\$560,572	\$74,436	\$0.00	\$635,008

**The NAD/USD conversation rate used was 15.11*

Consumption charges are related to the total amount of electricity consumed at the borehole scheme. A negative consumption charge indicates that the borehole scheme’s load demands were sufficiently met by the HRES, and the accumulated net-metering credits (sales) exceeded grid purchases. In this instance, the projected annual consumption charge savings were US\$560,572.

Demand charge is the charge for peak usage within a month. If the facility used a lot of power over short periods, the demand charges would comprise of a larger portion of the bill. If they use power at a more consistent rate throughout the month, the demand charges will generally comprise a smaller portion of the bill. Demand charges typically make up a significant portion of commercial and industrial consumers’ total electricity costs: typically, between 30 and 70%. The annual demand charge was reduced by 22.36% to US\$258,455, resulting in annual demand charge savings of US\$74,436. The projected total annual savings expected from implementing the proposed optimal HRES were US\$635,008.

4.3 Summary of Results

As the sensitivity variable, discount rate, was changed from 6.7%, to 10% and eventually 13%, different optimal system configurations and component sizes were determined. The key findings are summarised in Table 6

Table 6: Summary of key HRES economic metrics

Discount Rate	Optimal Configuration	Initial Capital	NPC	Total Annualized Savings	Plant Capacity Factor	LCOE
6.7%*	Onshore Wind, Solar PV, Storage and Grid	USD2.40 million	USD7.45 million	USD635,008	38.78%	USD0.0929 /kWh
10%	Onshore Wind, Solar PV, Storage and Grid	USD2.05 million	USD5.94 million	USD599,552	40.71%	USD0.105/ kWh
13%	Onshore Wind and Grid	USD1.73 million	USD4.99 million	USD556,807	42.77%	USD0.115/ kWh

The results also indicate that as the discount rate increases, lower capacity factor renewable technologies cost more over the project lifetime, even if they have a lower per kW installation cost (see also Table III.2). Hence when the optimization criterion was lowest NPC, solar PV’s installed capacity was reduced from 222kW to 101kW when the discount rate was increased from 6.7% to 10%. At a 13% discount rate, the lowest net present cost system excluded solar PV and storage all altogether. From Table III.2, solar PV has a cheaper per kW installation cost than onshore wind, however, because solar panels typically have capacity factors that are less than that of wind turbines, they cost more over the project lifetime because of lower electric energy production (relative to their installed capacity). Most importantly, the results show that even if the Namibian discount rate increases, the proposed HRESs and RES were economically feasible when implemented at the Berg Aukas borehole scheme. Each system’s LCOE was less than that of the utility grid i.e., USD 0.283 /kWh. Additionally, none the projected payback periods exceeded 5 years.

4.4 Discussion

The most notable benefit of implementing the HRESs at the borehole scheme was the reduced dependence on the grid for its operations with annual projected energy bill savings

of USD635,008, USD599,552 and USD556,807, at 6.7, 10 and 13 percent discount rates, respectively. Another noteworthy finding was that the net metering policy in Namibia also had an indirect influence on the LCOEs of the proposed HRESs.

The findings of this study correspond with those of other renewable energy projects in Africa. According to the International Renewable Energy Agency (IRENA), the African weighted-average LCOEs for onshore wind and utility-scale solar PV for 2020 was USD 0.055/kWh and USD 0.091/kWh, respectively [5]. The LCOE for the proposed HRES at the borehole scheme was determined to be USD 0.0929 /kWh. This suggests that the cost of implementing HRESs in Namibia for the purpose of powering borehole pumpstations closely resembles that of currently operational renewable energy projects across Africa. However, the Namibian results are even more impressive because at higher discount rates, the proposed systems' LCOEs were still lower than the utility grid i.e., USD 0.283/kWh.

The LCOE is widely used and is the industry standard for comparing energy related investments (ERIs) because of its simplicity, however, it has its limitations. The LCOE is most appropriately used for comparing similar systems in similar contexts. The LCOE is considered potentially misleading when used for direct comparisons between non-dispatchable sources (wind and solar) of energy and dispatchable sources because there is inherently more value in energy sources that can vary their output to follow load demand. This is a factor that is not adequately captured in the LCOE. Hence the use of the LCOE metric, alone, is not sufficient to definitively state that the HRESs are more economically favourable compared to the grid. The use of capacity expansion models has been cited as a more comprehensive approach to determining the economic competitiveness of energy systems [8].

5 Conclusions

This study had the main objective of assessing the techno-economic feasibility and sustainability of using HRESs to power borehole pumpstations in Namibia. The technologies used for the considered water supply scheme include solar PV, wind turbine, and battery storage. HOMER software was used to carry out the optimal sizing of the technologies and their performance evaluation. The annualized savings by implementing the proposed optimal HRES (at Namibia's average discount rate of 6.7%) was projected to be USD635,008). This translates to an annual energy cost reduction from the current utility bill of USD 807,743 down to USD172,735 i.e., 78.61% cost reduction. Additionally, the payback period of the initial capital investment was projected to be less 5 years. Furthermore, it was determined that implementing HRES to power the selected borehole scheme is economically feasible and sustainable

As with any engineering endeavour, it is important that solutions satisfy the triple bottom line of sustainability: economic sustainability, environmental sustainability, and social sustainability. The focus of this study was on economic sustainability, and it was determined that implementing HRESs to power the Berg Aukas borehole pump station is economically viable and sustainable. However, future work should investigate the social and environmental implications of introducing the produced solution. The impact of onshore wind turbines on the groundwater quality needs to be assessed. Wind turbines typically need deep foundations, and this could potentially compromise (contaminate) the aquifer's groundwater quality. These are important avenues for future investigations on using HRESs to power borehole schemes.

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